

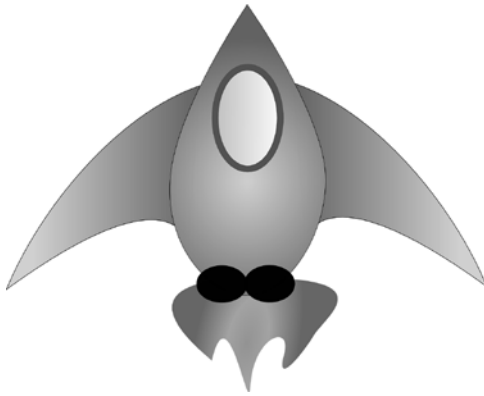


Figure 12-1: The ship for Alien Invasion

Creating the Ship Class

After choosing an image for the ship, we need to display it on the screen. To use our ship, we'll create a new ship module that will contain the class `Ship`. This class will manage most of the behavior of the player's ship:

```
ship.py import pygame

class Ship:
    """A class to manage the ship."""

    def __init__(self, ai_game):
        """Initialize the ship and set its starting position."""
        ❶ self.screen = ai_game.screen
        ❷ self.screen_rect = ai_game.screen.get_rect()

        # Load the ship image and get its rect.
        ❸ self.image = pygame.image.load('images/ship.bmp')
        self.rect = self.image.get_rect()

        # Start each new ship at the bottom center of the screen.
        ❹ self.rect.midbottom = self.screen_rect.midbottom

        ❺ def blitme(self):
            """Draw the ship at its current location."""
            self.screen.blit(self.image, self.rect)
```

Pygame is efficient because it lets you treat all game elements like rectangles (*rects*), even if they're not exactly shaped like rectangles. Treating an element as a rectangle is efficient because rectangles are simple geometric shapes. When Pygame needs to figure out whether two game elements have collided, for example, it can do this more quickly if it treats each object as a rectangle. This approach usually works well enough that no one playing the